In [1]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
```

In [2]:
```
df=pd.read_csv("C:/Users/Pranjal/Downloads/Social_Network_Ads.csv")
```

In [3]:
```
df
```

Out[3]:

	User ID	Gender	Age	EstimatedSalary	Purchased
0	15624510	Male	19	19000	0
1	15810944	Male	35	20000	0
2	15668575	Female	26	43000	0
3	15603246	Female	27	57000	0
4	15804002	Male	19	76000	0
...	...	...	...	...	...
395	15691863	Female	46	41000	1
396	15706071	Male	51	23000	1
397	15654296	Female	50	20000	1
398	15755018	Male	36	33000	0
399	15594041	Female	49	36000	1

400 rows × 5 columns

In [4]:
```
df.head()
```

Out[4]:

	User ID	Gender	Age	EstimatedSalary	Purchased
0	15624510	Male	19	19000	0
1	15810944	Male	35	20000	0
2	15668575	Female	26	43000	0
3	15603246	Female	27	57000	0
4	15804002	Male	19	76000	0

In [5]:
```
df.dropna()
```

Out[5]:

	User ID	Gender	Age	EstimatedSalary	Purchased
0	15624510	Male	19	19000	0
1	15810944	Male	35	20000	0
2	15668575	Female	26	43000	0
3	15603246	Female	27	57000	0
4	15804002	Male	19	76000	0
...	...	...	...	...	...
395	15691863	Female	46	41000	1
396	15706071	Male	51	23000	1
397	15654296	Female	50	20000	1
398	15755018	Male	36	33000	0
399	15594041	Female	49	36000	1

400 rows × 5 columns

In [6]:
```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 400 entries, 0 to 399
Data columns (total 5 columns):
 #   Column           Non-Null Count  Dtype 
---  ------           --------------  ----- 
 0   User ID          400 non-null    int64 
 1   Gender           400 non-null    object
 2   Age              400 non-null    int64 
 3   EstimatedSalary  400 non-null    int64 
 4   Purchased        400 non-null    int64 
dtypes: int64(4), object(1)
memory usage: 15.8+ KB
```

In [7]:
```
df=df.drop(["User ID"],axis=1)
```

In [8]:
```
from sklearn.preprocessing import LabelEncoder
le = LabelEncoder()
df['Gender'] = le.fit_transform(df['Gender'])
```

In [9]:
```
df
```

Out[9]:

	Gender	Age	EstimatedSalary	Purchased
0	1	19	19000	0
1	1	35	20000	0
2	0	26	43000	0
3	0	27	57000	0
4	1	19	76000	0
...	...	...	...	...
395	0	46	41000	1
396	1	51	23000	1
397	0	50	20000	1
398	1	36	33000	0
399	0	49	36000	1

400 rows × 4 columns

In [10]:
```
X = df[['Gender', 'Age', 'EstimatedSalary']]
y = df['Purchased']
```

In [11]:
```
from sklearn.model_selection import train_test_split

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.25, random_state=0
)
```

In [12]:
```
from sklearn.preprocessing import StandardScaler
sc = StandardScaler()
X_train = sc.fit_transform(X_train)
X_test = sc.transform(X_test)
```

In [13]:
```
from sklearn.linear_model import LogisticRegression

model = LogisticRegression()
model.fit(X_train, y_train)
```

Out[13]:

▾ LogisticRegression

▶ Parameters

In [14]:
```
y_pred = model.predict(X_test)
```

In [15]:
```
from sklearn.metrics import confusion_matrix, accuracy_score

cm = confusion_matrix(y_test, y_pred)
print("Accuracy:", accuracy_score(y_test, y_pred))
```

```
Accuracy: 0.9
```

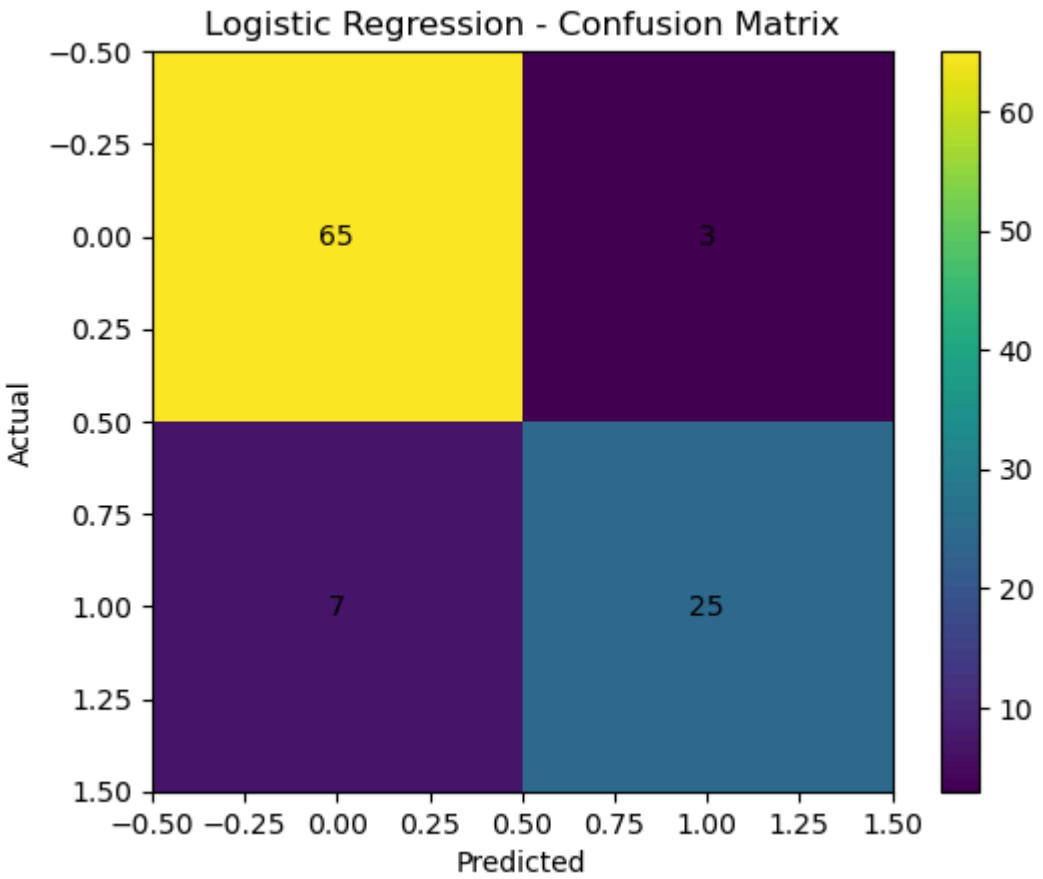
In [16]:
```
print("Confusion Matrix:\n", cm)
```

```
Confusion Matrix:
 [[65  3]
 [ 7 25]]
```

In [17]:
```
from sklearn.metrics import accuracy_score, precision_score, recall_score, f1_sc
print("Accuracy:", accuracy_score(y_test, y_pred))
print("Precision:", precision_score(y_test, y_pred))
print("Recall:", recall_score(y_test, y_pred))
print("F1 Score:", f1_score(y_test, y_pred))
```

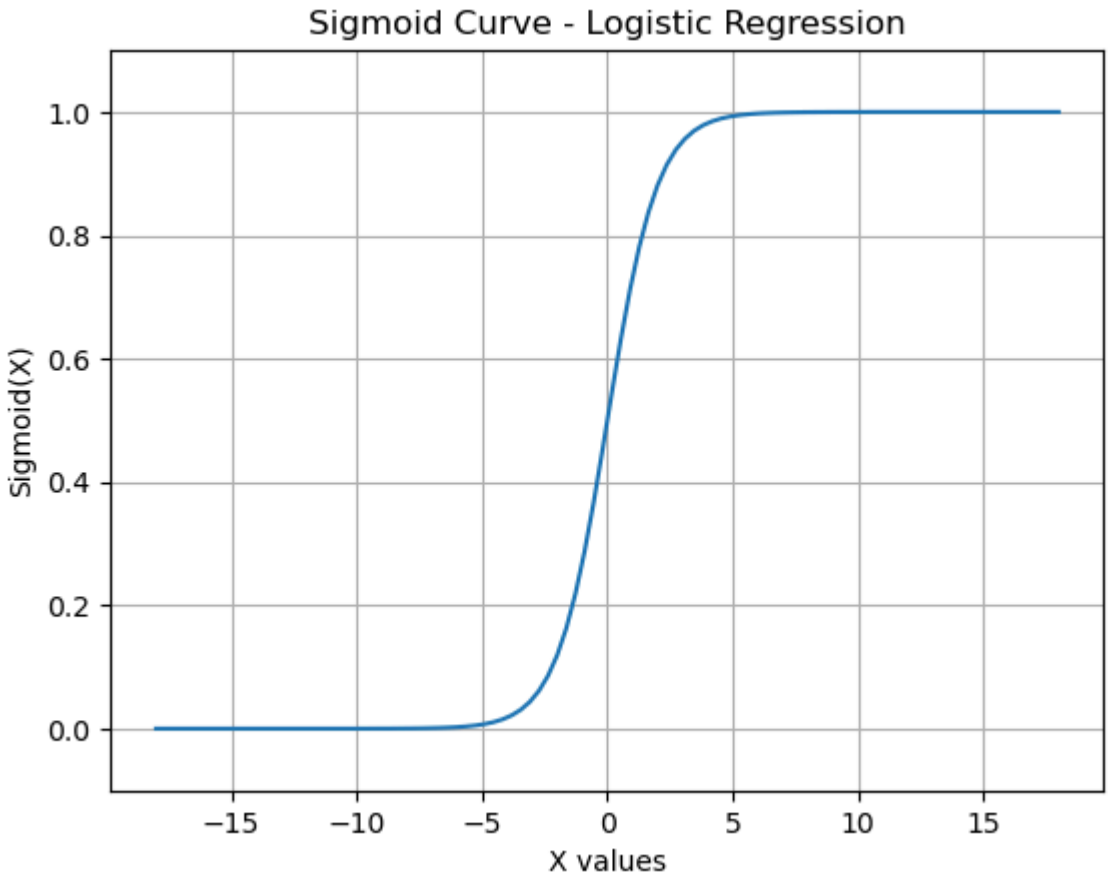
```
Accuracy: 0.9
Precision: 0.8928571428571429
Recall: 0.78125
F1 Score: 0.8333333333333334
```

In [18]:
```
cm = confusion_matrix(y_test, y_pred)
plt.figure()
plt.imshow(cm)
plt.title("Logistic Regression - Confusion Matrix")
plt.xlabel("Predicted")
plt.ylabel("Actual")
for i in range(len(cm)):
    for j in range(len(cm[0])):
        plt.text(j, i, cm[i][j], ha='center', va='center')
plt.colorbar()
plt.show()
```



In [19]:
```
x = np.linspace(-18, 18, 100)
y = 1 / (1 + np.exp(-x))
plt.figure()
plt.plot(x, y)
```

```
plt.title("Sigmoid Curve - Logistic Regression")
plt.xlabel("X values")
plt.ylabel("Sigmoid(X)")
plt.ylim(-0.1, 1.1)
plt.grid()
plt.show()
```



In [ ]: